

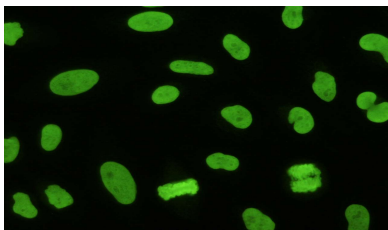
Derbyshire Shared Care Pathology Guidelines

Antinuclear antibody (ANA) testing

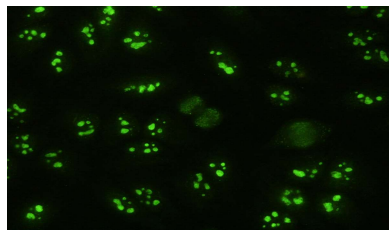
- ANA testing is helpful in patients with possible connective tissue disease
- Positive ANA can occur in infections and in up to 15% of healthy individuals
- Positive results in patients with vague non-specific symptoms may result in unnecessary further testing and worry
- Negative ANA makes SLE unlikely, but does not exclude many other inflammatory disorders

What is a positive ANA?

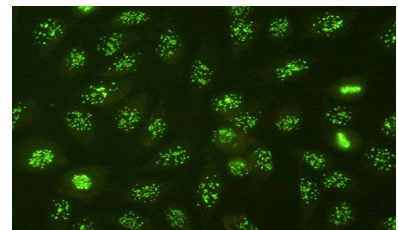
Antinuclear antibodies are autoantibodies directed against the cell nucleus. They are identified by applying patient serum to cells fixed on a slide and inspecting these under a microscope. If the patient's IgG antibodies bind to the nucleus, that is a positive ANA.



Homogenous ANA



Nucleolar ANA



Centromere ANA

The test is not fully automated, so it takes 2-3 days to issue a result. Follow-on tests can take longer.

Who should have an ANA test?

ANA is a useful screening test in patients who may have SLE or other connective tissue diseases (CTDs). They are also used to classify autoimmune hepatitis and in juvenile arthritis.

However, because ANA can be positive in other inflammatory diseases (e.g. thyroiditis), infections (viral and chronic bacterial), with some drugs and even in healthy individuals, they must be requested with discretion. Positive results in patients with vague non-specific symptoms may result in unnecessary further testing and worry.

Scenario 1: A 58-year-old man with aches and pains for several weeks

Result: ANA+ homogenous 1:160

Interpretation: Unhelpful (unless there are other clinical or laboratory features of CTD)

The ANA result (whether positive or negative) did not change the likely diagnosis

Scenario 2: A 32-year-old woman with arthralgia, hair loss and a photosensitive rash

Result: ANA+ speckled 1:160

Interpretation: Needs further investigation for SLE (or other CTD)

The likelihood of the patient having a connective tissue disease was quite high before the test was requested. The positive result increased the likelihood of SLE.

Summary guidance on testing and interpretation of ANAs

SLE and other CTDs are rare. A GP with 3000 patients will have about 3 SLE patients on their books and will diagnose a new patient about every 7 years.

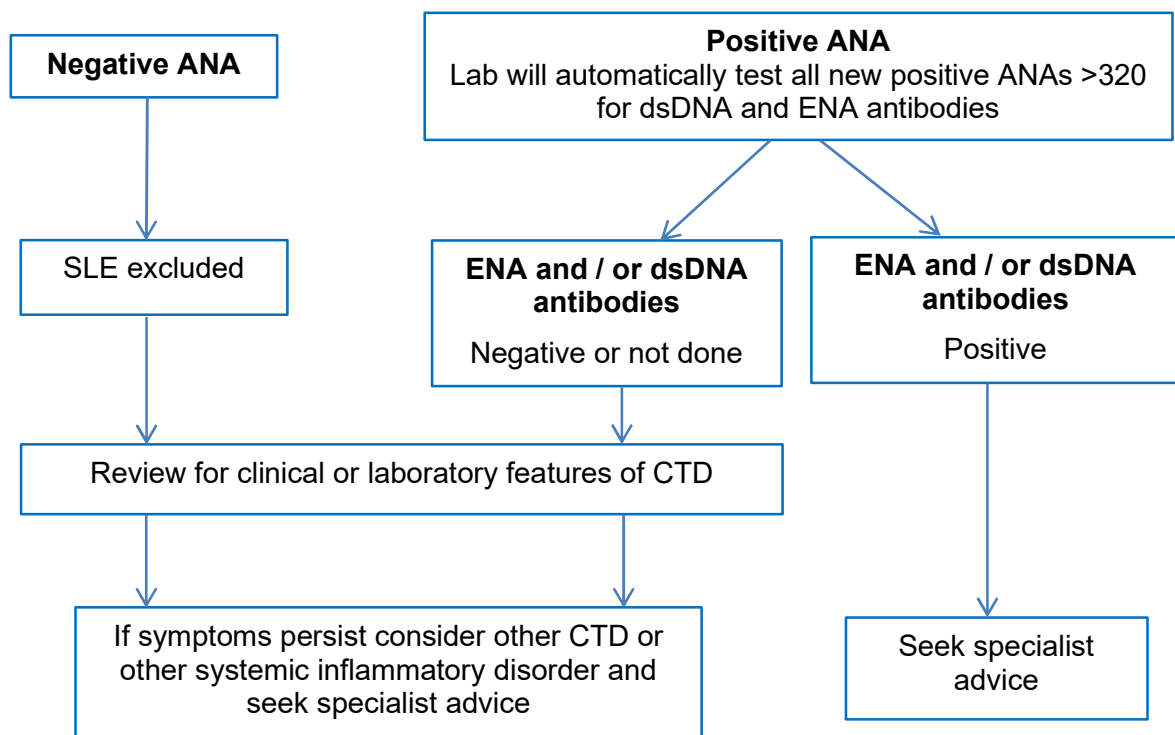
SLE is commoner in younger females, and patients of Afro-Caribbean and Asian origin are at higher risk.

Commonest symptoms of patients with SLE or Connective Tissue Disease (CTD)

Joint pains with synovitis, stiffness	Severe oral ulceration
Raynaud's phenomenon	Ocular inflammation
Rashes and photosensitivity	Unexplained thrombosis or pregnancy loss
Pleuritic chest pain, pericarditis	Thickening of the skin
Mild fever	Neurological symptoms
Dry eyes and mouth	Muscle weakness or tenderness
	Proteinuria or haematuria

Some laboratory features of SLE or CTD

Raised ESR / CRP (*CRP often normal in SLE*)
Leucopenia / lymphopenia
Anaemia
Thrombocytopenia
Renal impairment
Raised creatine kinase
Blood or protein on urinalysis
Low C3 and/or C4
Anti-phospholipid / cardiolipin antibodies
Lupus anticoagulant



Which diseases typically have a positive ANA?

Disease	% ANA-Positive
SLE	>95
Diffuse systemic sclerosis	85–95
Sjögren's syndrome	50–60
Autoimmune myositis	40–60
Mixed Connective Tissue Disease	90-100

What is the significance of the pattern and titre?

The pattern of ANA can be helpful in identifying a specific disease. Homogenous or speckled ANA can occur in SLE (>95% of patients), but is quite common in healthy individuals (up to 15%). Centromere antibodies are highly suggestive of the CREST variant of systemic sclerosis. Most patterns do not have well-defined disease associations.

The higher the titre, the more likely it is that the patient has an associated condition. However, even strongly positive ANAs (>1:1280) can occur transiently in infections, or even in healthy individuals.

If ANA is negative, have I excluded inflammatory diseases?

Many important autoimmune and inflammatory diseases, which may have similar clinical features, are typically ANA-negative. These include many patients with conditions that we think of as having a positive ANA.

Vasculitis	Juvenile chronic arthritis
Temporal arteritis / Polymyalgia rheumatica	Polymyositis / dermatomyositis
Rheumatoid arthritis	Sjogren's syndrome
Psoriatic and other spondyloarthritides	

If these are suspected clinically, then seek specialist advice on further specific testing.

Follow-on testing – ENA and dsDNA antibodies

The cell nucleus contains a large number of antigens, any of which can be the target of the antinuclear antibodies. Double-stranded DNA (dsDNA) and RNA are examples. Others, known as extractable nuclear antigens (ENAs) are known by specific names.

Antibody Name	Disease association
dsDNA	SLE
Ro (SSA)	Lupus (cutaneous or systemic), Sjogren's, neonatal heart block
La (SSB)	Lupus (cutaneous or systemic), Sjogren's
RNP	SLE
Sm	SLE
Jo-1	Polymyositis
Scl-70	Scleroderma

The lab will automatically test all new positive ANAs >320 for dsDNA and ENA antibodies. These only need to be specifically requested if the disease (e.g. Sjogren's or polymyositis) is strongly suspected, but the ANA is negative.

While positive ENA antibodies are far more likely to indicate disease than a positive ANA alone, they are not diagnostic in isolation. ENA antibodies are also absent in a proportion of patients with these diseases, so a negative test does not exclude the diagnosis.

Patients with persistent symptoms of systemic inflammatory diseases should be discussed or referred regardless of their ANA, ENA or dsDNA test results.

Repeat testing and monitoring of ANA+ patients

It is not usually helpful to repeat a positive ANA.

A transient ANA (such as may occur in infection) will take many months to disappear. Re-testing after a year may help to exclude some ANAs of doubtful clinical significance, but this is not necessary if the symptoms have resolved. A persisting ANA is not evidence of disease in the absence of relevant clinical features.

Repeat ANA testing is of *no value* in monitoring diseases in patients with associated diseases such as SLE or autoimmune hepatitis.

Contacts

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References

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